

Late Delivery Prediction using DataCo Supply Chain Dataset

Submitted By

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Project Title: Late Delivery Prediction using DataCo Supply Chain Dataset

1. Introduction

Supply chain management plays a crucial role in ensuring that products are delivered to customers on time. Late deliveries can lead to customer dissatisfaction, increased operational costs, and loss of business reputation. The objective of this project is to build a machine learning model that predicts whether an order will be delivered late or on time using the DataCo Supply Chain dataset.

The project involves data cleaning, feature engineering, model building, evaluation, and feature importance analysis.

2. Dataset Description

Dataset Name: DataCoSupplyChainDataset.csv

The dataset contains information related to customers, orders, products, shipping details, and delivery status.

Number of Records: 180,519

Number of Features: 53 (before preprocessing)

The dataset includes variables such as:

- Customer information
 - Product details
 - Sales and profit information
 - Shipping mode and delivery details
 - Order date and shipping date
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3. Question 1: Data Quality Assessment and Data Cleaning

Issues Identified

1. Completely Missing Column

- Product Description contained 100% missing values.
- Action Taken: Dropped the column because it did not contain any useful information.

2. Highly Missing Column

- Order Zipcode contained approximately 86% missing values.
- Action Taken: Dropped the column because excessive missing values make the feature unreliable.

3. Missing Values

- Customer Lname had a few missing values.
- Filled with "Unknown".
- Customer Zipcode had a few missing values.
- Filled using the mode because zip codes are categorical in nature.

4. Redundant Columns

The following columns were removed because they were identifiers or sensitive information and did not contribute to prediction:

- Customer Email
- Customer Password
- Customer Fname
- Customer Street
- Order Id
- Order Item Id

5. Date Conversion

Order date and shipping date columns were converted into datetime format to enable feature extraction.

After preprocessing, the dataset became suitable for machine learning analysis.

4. Question 2: Target Variable Construction

The objective is to predict whether an order is delivered late.

Target variable definition:

- 0 = On-Time Delivery
- 1 = Late Delivery

The existing Late_delivery_risk information was used as the delivery outcome indicator after removing it from the feature set to avoid data leakage.

Class Distribution

(Add your output table here)

Class	Description	Count	Percentage
0	On-Time Delivery	77119	57.28%
1	Late Delivery	103400	42.72%

5. Question 3: Feature Engineering

The following features were engineered:

1. Order_Month

Extracted from order date.

Reason:

Different months may experience seasonal demand and shipping delays.

2. Order_DayOfWeek

Extracted from order date.

Reason:

Orders placed on weekends or specific weekdays may have different delivery patterns.

3. Order_Hour

Extracted from order date.

Reason:

Orders placed late in the day may experience processing delays.

4. Discount_Percentage

Formula:

Discount Percentage = (Order Item Discount ÷ Sales) × 100

Reason:

Higher discounts may generate large order volumes, increasing the possibility of delayed deliveries.

These engineered features provide additional information without directly revealing the target outcome.

6. Question 4: Model Building and Evaluation

Data Split

Training Set: 80%

Testing Set: 20%

Stratified sampling was used to preserve the class distribution.

Model Used

Random Forest Classifier

Evaluation Metrics

Metric	Value
Accuracy	85.29%
Precision	92.22%
Recall	81.17%
F1-Score	86.34%
ROC-AUC	94.38%

Interpretation

The model achieved high precision and a strong ROC-AUC score, indicating excellent capability in distinguishing late deliveries from on-time deliveries.

7. Question 5: Model Improvement

Techniques attempted:

- Hyperparameter Tuning
- Class Imbalance Assessment
- Feature Selection

The baseline Random Forest model already demonstrated strong performance, and further tuning resulted in only marginal improvements.

Baseline vs Improved Model

Metric	Baseline	Improved
Accuracy	85.29%	85.29%
Precision	92.22%	92.22%
Recall	81.17%	81.17%
F1-Score	86.34%	86.34%
ROC-AUC	94.38%	94.38%

Final Model Selection

The baseline Random Forest model was selected as the final model because it provided excellent performance and additional tuning did not significantly improve the results.

8. Question 6: Top 5 Influential Features

Feature importance scores from the Random Forest model were used to identify the most influential predictors.

Top 5 Features

(Add your actual output table here)

Rank	Feature	Importance Score
1	Shipping Mode	0.241985
2	Latitude	0.040271
3	Order City	0.039977
4	Order Customer Id	0.038750
5	Customer Id	0.038718

Interpretation

The top features significantly influenced the model's predictions and played an important role in identifying whether an order would be delivered late or on time.

9. Workflow

Dataset Collection
↓
Data Cleaning and Preprocessing
↓
Target Variable Creation
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Feature Engineering
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Data Encoding
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Train-Test Split
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Random Forest Model Training
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Model Evaluation
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Model Improvement
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Feature Importance Analysis
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Final Conclusion

10. Conclusion

This project successfully developed a machine learning model for predicting late deliveries using the DataCo Supply Chain dataset. Data preprocessing and feature engineering improved the quality of the dataset, and the Random Forest model achieved strong predictive performance with an Accuracy of 85.29% and ROC-AUC of 94.38%.

Feature importance analysis identified the key factors influencing delivery performance. The developed model can help organizations anticipate delivery delays and support better supply chain decision-making.